

Twenty (simple) questions — moderate

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For every distribution μ on $[n]$, we can find a prefix code whose average codeword length is less than $H(\mu) + 1$ (where we use base 2 logarithms in the definition of entropy), known as Shannon–Fano coding. We can think of the prefix code as a strategy for determining a random $x \sim \mu$ using Yes/No questions which uses fewer than $H(\mu) + 1$ questions in average.

What happens if we restrict the set of allowed questions? For questions of the form “ $x < m$ ” (corresponding to binary search trees), the bound becomes $H(\mu) + 2$.

The paper “Twenty (simple) questions” asks whether we can recover the bound $H(\mu) + 1$ using a small inventory of questions, and answers this in the affirmative: it suffices to consider questions of the form “ $x < m$ ” and “ $x = m$ ”.

What happens when we consider questions with at most c possible answers for some $c > 2$? Shannon–Fano coding still works: there is a strategy that uses fewer than $H_c(\mu) + 1$ questions in average, where H_c is base c entropy.

The natural generalization of “ $x < m$ ” to the c -ary setting is to consider partitions of $[n]$ into at most c intervals. This results in the bound $H_c(\mu) + \log_c 2$ (one can see this using the Gilbert–Moore algorithm, for example).

Problem 1. *Find a small and natural inventory of c -way questions $\mathcal{I}_n$ such that for each μ there is a strategy using only questions in $\mathcal{I}_n$ which asks fewer than $H_c(\mu) + 1$ questions on average.*

A natural guess is to generalize the inventory of “Twenty (simple) questions” to the c -way setting.